

Tech Migration – From Ericsson GSC China (AWS Cloud)

Client: Zain Iraq

Employer: Ericsson (Performance Team)

Location: Ericsson Lahore Office (On-site)

Project Background

The project aimed to facilitate the handover of Ericsson GSC China's manual workflows for processing Zain Iraq's 2G/3G/4G/5G network performance KPIs to the Pakistan based Ericsson Performance Team by developing an automated, cloud-native reporting platform on AWS.

Our performance team containerized legacy Python scripts previously run on individual laptops to automate XML parsing, Excel reporting, and performance visualization.

Phase I: AWS EC2 Deployment & Network Collaboration

The first phase focused on deploying the legacy Python scripts on AWS EC2 instances, transitioning from the previous manual process performed by the Ericsson GSC China team on their laptops. During this phase, we collaborated with the Ericsson India team to resolve ongoing network issues related to AWS Active Directory, which was used for securely placing XML files uploaded by the Iraq team (each XML representing network towers).

Phase II: Containerization & Kubernetes Orchestration on AWS EKS

In the second phase, the workloads were fully containerized and deployed on an AWS EKS cluster, using GitOps practices to enable continuous delivery. Along with two team members, we handled infrastructure automation and set up CI/CD pipelines throughout phase II.

Raw XML data was transformed into CSV files and securely ingested via VPN into AWS S3, where Dockerized Python workloads running on Kubernetes processed the data. The processed results were stored in AWS RDS (MSSQL), with reports generated as Excel visualizations and interactive HTML dashboards, securely delivered through an Application Load Balancer.

Infrastructure was codified using Terraform for AWS provisioning and Ansible for EC2/EKS node configuration. CI/CD was implemented using GitHub Actions for image builds and testing, with ArgoCD for automated GitOps based deployment. Additional deliverables included Prometheus + Grafana monitoring, secure IAM integration with Active Directory, and packaging internal services for long-term use.

Throughout the journey, I documented my learning and DevOps practices in this GitHub repository: <https://github.com/khaleel-git/DevOPS>